



VYBRION CIVIL TECH ACADEMY

COURSE SYLLABUS

Software & Technical Courses in Civil Engineering

26 Industry-Aligned Courses

12 Software Courses • 14 Technical Courses

Every course includes a capstone project, portfolio deliverable, and certification.

VYBRION
— CIVIL TECH ACADEMY —
Learn Beyond

How This Syllabus Is Structured

Every course in this catalog is built on the same academic standard, so learners always know what to expect and employers can trust the depth behind every certificate.

Session Standard

- 1 Session = 2 Hours (1.5 Hrs Teaching/Demonstration + 0.5 Hr Practice)
- 3 Sessions per Week = 6 Hours per Week

Software Course Path

- Basic → Intermediate → Advanced → Practical Project

Technical Course Path

- Fundamentals → Analysis → Design → Code-Based Problems → Practical Application

Every Course Includes

- A course overview and target audience
- Prerequisites, so learners self-select correctly
- Clear, measurable learning outcomes
- A detailed week-by-week curriculum
- The exact tools/software and reference codes used
- A capstone project — a real, portfolio-worthy deliverable
- Certification on completion

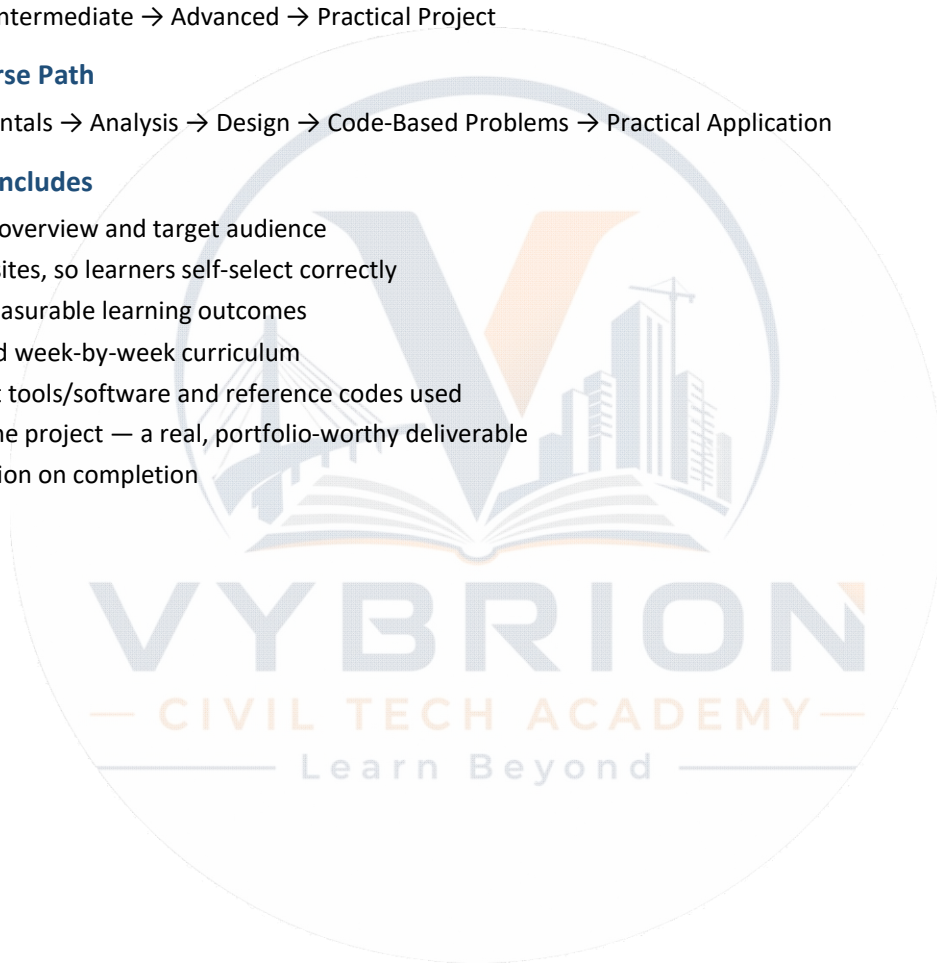


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1. Software Courses

Design, drafting, modeling, BIM & project-control tools used across the civil/construction industry

1. AutoCAD

Software Course | Duration: 6 Weeks

A hands-on 2D drafting course that takes learners from blank-canvas basics to producing complete, print-ready construction drawing sets used in real civil and architectural offices.

Who Should Enroll: Civil/Architecture students, diploma holders, and working professionals who need to read, draft, or edit construction drawings.

Prerequisites

- Basic computer literacy
- Elementary understanding of building plans is helpful but not mandatory

Learning Outcomes

- Draft accurate 2D plans, elevations, and sections to scale
- Apply layers, dimensioning, and annotation standards used in professional offices
- Build and reuse blocks, templates, and title blocks for faster drafting
- Set up print-ready layouts and sheet sets for site and client submission
- Independently produce a full working drawing set for a small building

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Interface & Draw Tools	– Workspace, coordinate systems, and units setup – Line, circle, arc, polyline, rectangle commands – Precision drafting with object snaps and ortho/polar tracking
Week 2	Modify & Organize	– Trim, extend, offset, fillet, chamfer, array, mirror – Layers, line types, colors, and object properties – Practice: converting a hand sketch into a scaled digital drawing
Week 3	Annotation & Reuse	– Dimension styles and annotation standards – Creating and inserting blocks with attributes – Hatching, text styles, and drawing plan/elevation/section views
Week 4	Sheet Setup & Output	– Layouts, viewports, and scale management – Title blocks, sheet sets, and plot/print standards – Xrefs for coordinating multi-drawing projects
Week 5	Working Drawing Practice	– Drafting full plan, elevation, and section sheets for a sample building – Detail drawings: staircase, foundation, door/window schedules
Week 6	Capstone Project	– Complete, print-ready drawing set for a residential/commercial building – Peer review and correction round – Final submission in PDF and DWG

Tools & Reference Standards

- AutoCAD 2D/2D-Draft & Annotate workspace
- DWG/DXF file management
- PDF plotting

Capstone Project

A full working-drawing package (plan, elevation, section, and details) for a residential or small commercial building, submitted print-ready.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, along with a portfolio-ready project file that can be showcased to employers or clients.

2. STAAD.Pro

Software Course | Duration: 8 Weeks

A structural analysis and design course covering modeling, loading, code-based analysis, and RC/steel member design for multi-storey buildings using STAAD.Pro — the industry-standard tool for structural consultants.

Who Should Enroll: Civil engineering students, structural design trainees, and site/design engineers moving into structural analysis roles.

Prerequisites

- Basic structural analysis / engineering mechanics
- Familiarity with IS 456 and IS 800 is helpful

Learning Outcomes

- Model 3D framed structures with correct geometry, sections, and supports
- Apply dead, live, wind, and seismic loads as per IS 875 and IS 1893
- Interpret analysis results — BM, SF, deflection, and reactions
- Design RC beams/columns/footings and steel members within the software
- Produce a complete design report for a real building frame

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Modeling Basics	– Interface, units, and coordinate systems – Creating nodes, beams, and plate geometry – Assigning material and section properties
Week 2	Supports & Geometry Practice	– Support conditions: fixed, pinned, roller, spring – Generating multi-storey geometry using repeat/translate tools
Week 3–4	Loading & Combinations	– Dead, live, wind (IS 875) and seismic (IS 1893) load definition – Load combinations and load generation for lateral cases – Running analysis and diagnosing model/analysis errors
Week 5	Result Interpretation	– Reading BM, SF, deflection, and reaction outputs – Cross-checking software output against manual estimates
Week 6–7	Design Modules	– RC design of beams, columns, and footings (IS 456) – Steel member design (IS 800) – Response spectrum / dynamic analysis basics

Week	Focus	Topics Covered
Week 8	Capstone Project	– End-to-end analysis and design of a multi-storey building frame – Design report generation and BBS-ready output – Review session and Q&A on real project queries

Tools & Reference Standards

- STAAD.Pro V8i/CONNECT Edition
- IS 456, IS 800, IS 875, IS 1893 codebooks

Capstone Project

Full structural analysis and design of a multi-storey RC/steel building frame, with a generated design report.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, along with a portfolio-ready project file that can be showcased to employers or clients.

3. CSI SAFE

Software Course | Duration: 5 Weeks

A focused course on slab and foundation design using CSI SAFE, covering RC and post-tensioned slab systems, punching shear, and mat/raft foundation design commonly required in high-rise and industrial projects.

Who Should Enroll: Structural design engineers and STAAD/ETABS users who want to specialize in slab and foundation design.

Prerequisites

- Basic RCC design knowledge
- Familiarity with STAAD.Pro or ETABS is an advantage

Learning Outcomes

- Model slab, beam, and foundation systems with correct grid and property setup
- Apply realistic loading for flat slabs, PT slabs, and rafts
- Run and interpret meshed finite-element slab analysis
- Perform punching shear checks and IS 456-based slab design
- Design a mat/raft foundation with reinforcement detailing

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Modeling Fundamentals	– Interface overview, grid and structural layout setup – Slab, beam, and column property definitions
Week 2	Loading & Slab Types	– Load assignment: dead, live, finishes, partition loads – Modeling conventional RC slabs vs post-tensioned slabs
Week 3	Analysis	– Mesh generation strategy and analysis run – Reading moment/shear contours and deflection results
Week 4	Design Checks	– Punching shear checks around columns – Slab and mat foundation design per IS 456 – Deflection and crack-width verification

Week	Focus	Topics Covered
Week 5	Capstone Project	– Design of a flat-slab or raft foundation system for a sample building – Design report with reinforcement detailing output

Tools & Reference Standards

- CSI SAFE
- IS 456 codebook

Capstone Project

Design of a flat slab / raft foundation for a sample multi-storey building, including a full design report.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, along with a portfolio-ready project file that can be showcased to employers or clients.

4. ETABS

Software Course | Duration: 7 Weeks

A complete 3D building analysis and design course using ETABS — covering seismic and wind analysis, storey drift and irregularity checks, and RC/steel frame design for multi-storey structures.

Who Should Enroll: Structural engineers, design trainees, and STAAD.Pro users transitioning into 3D building analysis for mid-to-high-rise projects.

Prerequisites

- Basic structural analysis and RC design concepts
- STAAD.Pro exposure is helpful but not required

Learning Outcomes

- Build a complete 3D building model with correct storey and diaphragm data
- Apply gravity, wind, and seismic loads per Indian codes
- Run response spectrum and equivalent static seismic analysis
- Check storey drift, base shear, and structural irregularities
- Design RC/steel frame members and produce a design report

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Model Setup	– Interface, grids, and storey data – Modeling beams, columns, slabs, and shear walls
Week 2	Properties & Sections	– Material and section property libraries – Assigning releases, offsets, and rigid diaphragms
Week 3–4	Loading	– Gravity, wind, and seismic load application – Load combinations per IS 875 & IS 1893 – Diaphragm behavior: rigid vs semi-rigid modeling
Week 5	Seismic Analysis	– Response spectrum and equivalent static analysis – Storey drift, base shear, and irregularity checks

Week	Focus	Topics Covered
Week 6	Design Modules	- Concrete and steel frame design within ETABS - Cross-checking design output against manual calculations
Week 7	Capstone Project	- Full 3D seismic analysis and design of a multi-storey RC building - Final design report and performance summary

Tools & Reference Standards

- ETABS (CSI)
- IS 1893, IS 875, IS 456, IS 800 codebooks

Capstone Project

Complete 3D seismic analysis, design, and performance evaluation of a multi-storey RC building.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, along with a portfolio-ready project file that can be showcased to employers or clients.

5. RCDC

Software Course | Duration: 4 Weeks

A production-focused course on RCDC (RC Design & Comprehensive detailing) for converting analysis models from STAAD/ETABS into fully detailed, construction-ready RC drawings and bar bending schedules.

Who Should Enroll: Design engineers and detailers who need to speed up RC detailing and BBS generation for real projects.

Prerequisites

- Working knowledge of STAAD.Pro or ETABS
- Basic RCC design concepts

Learning Outcomes

- Import and organize analysis models from STAAD/ETABS into RCDC
- Configure design groups and code-based design parameters
- Generate detailed reinforcement drawings for beams, columns, and slabs
- Produce accurate bar bending schedules (BBS) for site use
- Optimize designs and resolve detailing clashes

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Setup & Import	- Interface overview and STAAD/ETABS model import - Understanding auto-generated design groups
Week 2	Design Configuration	- Beam, column, and slab design settings - Design code selection and parameter customization
Week 3	Detailing	- Detailed reinforcement detailing for members - Bar bending schedule (BBS) generation - Design optimization and clash checks

Week	Focus	Topics Covered
Week 4	Capstone Project	– Complete RC detailing package for a sample structure – Final BBS and drawing export ready for site use

Tools & Reference Standards

- RCDC
- STAAD.Pro/ETABS model interoperability
- IS 456, SP 34 detailing standards

Capstone Project

Full RC detailing and BBS package for a sample building, exported in a site-ready format.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, along with a portfolio-ready project file that can be showcased to employers or clients.

6. IDEA StatiCa

Software Course | Duration: 4 Weeks

A specialized steel connection design course using IDEA StatiCa's CBFEM technology, covering bolted/welded joints, base plates, and stiffener design for real industrial and commercial steel structures.

Who Should Enroll: Steel structure design engineers and STAAD/ETABS users who need to design and verify complex steel connections.

Prerequisites

- Basic steel design knowledge (IS 800)
- Familiarity with STAAD.Pro or ETABS frame models

Learning Outcomes

- Import frame models and identify critical connections for design
- Design bolted and welded steel connections
- Design column base plates and anchor bolt groups
- Run CBFEM code-checks on complex multi-member joints
- Classify joint stiffness and design stiffeners where required

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Connection Modeling	– Interface overview and connection modeling basics – Importing frame models from STAAD/ETABS
Week 2	Bolted & Welded Joints	– Steel connection design: bolted and welded joints – Base plate and anchor bolt design
Week 3	Advanced Checks	– Code-check (CBFEM) analysis of complex joints – Stiffener design and joint stiffness classification
Week 4	Capstone Project	– Design and verification of steel connections for a sample industrial structure – Final connection design report

Tools & Reference Standards

- IDEA StatiCa Connection
- IS 800 codebook

Capstone Project

Design and CBFEM verification of steel connections for a sample industrial shed/structure.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, along with a portfolio-ready project file that can be showcased to employers or clients.

7. Revit Architecture

Software Course | Duration: 7 Weeks

A complete BIM authoring course for architecture — from parametric modeling and families to full construction documentation — preparing learners to deliver coordinated architectural BIM models for real projects.

Who Should Enroll: Architecture students, draftspersons, and CAD professionals moving from 2D drafting to 3D BIM workflows.

Prerequisites

- Basic understanding of building plans
- AutoCAD experience is helpful but not mandatory

Learning Outcomes

- Build a parametric architectural model using levels, grids, and walls
- Create and customize families for doors, windows, and furniture
- Model stairs, railings, roofs, and site context
- Generate rooms, area schedules, and quantity take-offs
- Produce a full, sheet-ready construction documentation set

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Project Setup	– Interface, templates, and project setup – Levels, grids, walls, doors, and windows
Week 2	Building Elements	– Floor and roof modeling – Practice: modeling a two-storey structure shell
Week 3–4	Families & Components	– Creating and editing custom families – Stairs, railings, and site modeling – Rooms, areas, and schedules
Week 5	Documentation	– Sheet setup, annotation, and dimensioning – View templates and drawing standards
Week 6	Visualization & Collaboration	– Rendering and visualization basics – Worksharing fundamentals for team collaboration
Week 7	Capstone Project	– Full architectural BIM model of a residential/commercial building – Complete construction documentation set (plans, elevations, sections, sheets)

Tools & Reference Standards

- Autodesk Revit (Architecture)
- Family Editor
- Revit Schedules

Capstone Project

Full architectural BIM model with a coordinated construction documentation set for a real-scale building.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, along with a portfolio-ready project file that can be showcased to employers or clients.

8. Revit Structure

Software Course | Duration: 7 Weeks

A structural BIM course covering modeling of columns, beams, footings and rebar, coordination with architectural models, and clash-checked documentation for multi-storey buildings.

Who Should Enroll: Structural draftspersons and design engineers moving from 2D detailing into coordinated structural BIM.

Prerequisites

- Basic RCC design/detailing knowledge
- Revit Architecture familiarity is an advantage

Learning Outcomes

- Model structural columns, beams, footings, and slabs accurately
- Create rebar models and reinforcement detailing in 3D
- Link and coordinate structural models with architectural models
- Run basic clash detection and resolve coordination issues
- Extract structural schedules and quantities directly from the model

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Structural Modeling Basics	– Interface, structural templates, and grid setup – Modeling columns, beams, and footings
Week 2	Slabs & Framing	– Structural slab and framing modeling – Applying structural materials and analytical settings
Week 3–4	Rebar & Detailing	– Rebar modeling and reinforcement detailing – Structural families and connection details
Week 5	Coordination	– Linking architectural and structural models – Clash detection basics and issue resolution
Week 6	Schedules & Documentation	– Structural schedules and quantity extraction – Sheet setup for structural drawing sets
Week 7	Capstone Project	– Complete structural BIM model for a multi-storey building – Coordinated documentation set with rebar detailing

Tools & Reference Standards

- Autodesk Revit (Structure)
- Rebar modeling tools
- Navisworks (basic interoperability)

Capstone Project

Coordinated structural BIM model with rebar detailing and a full structural drawing set.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, along with a portfolio-ready project file that can be showcased to employers or clients.

9. SketchUp

Software Course | Duration: 5 Weeks

A fast, visual 3D modeling course for architects and designers — covering conceptual massing, materials, and site context modeling to communicate design ideas quickly and clearly.

Who Should Enroll: Architecture and interior design students, and professionals who need quick conceptual 3D modeling skills.

Prerequisites

- No prior 3D modeling experience required

Learning Outcomes

- Build accurate 3D massing models using push-pull modeling
- Organize models using groups, components, and layers
- Apply materials, textures, and basic terrain/site context
- Set up scenes and walkthroughs for design presentation
- Produce a presentation-ready conceptual model of a building

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Modeling Basics	– Interface, navigation, and drawing tools – Basic 3D massing and push-pull modeling
Week 2	Organization	– Groups, components, and layers – Editing and reusing components efficiently
Week 3	Materials & Context	– Materials, textures, and basic terrain modeling – Adding site context: roads, landscaping, neighboring structures
Week 4	Presentation Tools	– Scenes, animation, and walkthroughs – Plugin/extension basics for enhanced rendering
Week 5	Capstone Project	– 3D conceptual model of a building with site context – Rendered views and a walkthrough presentation

Tools & Reference Standards

- SketchUp Pro
- V-Ray or equivalent rendering plugin (introductory)

Capstone Project

A presentation-ready 3D conceptual model with site context, rendered views, and a walkthrough.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, along with a portfolio-ready project file that can be showcased to employers or clients.

10. Lumion

Software Course | Duration: 4 Weeks

A photorealistic rendering and animation course that turns 3D architectural models into cinematic stills and walkthroughs — a key skill for client presentations and marketing.

Who Should Enroll: Architects, interior designers, and visualization artists who want to produce high-quality renders and animations.

Prerequisites

- A basic 3D model from SketchUp, Revit, or similar software

Learning Outcomes

- Import and prepare 3D models for rendering in Lumion
- Build realistic environments with landscape, lighting, and weather effects
- Populate scenes with context: people, vegetation, and vehicles
- Create camera paths and animation sequences
- Produce final photorealistic stills and walkthrough videos

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Import & Environment	– Interface overview and importing 3D models – Landscape, terrain, and environment setup
Week 2	Materials & Context	– Materials, lighting, and weather effects – Adding people, vegetation, and vehicles for realism
Week 3	Animation	– Camera paths and animation sequences – Rendering settings for stills vs walkthrough video
Week 4	Capstone Project	– Photorealistic render set for an imported building model – Final animated walkthrough video

Tools & Reference Standards

- Lumion
- Compatible 3D model formats (SketchUp/Revit exports)

Capstone Project

A photorealistic render package and animated walkthrough video for a client-style presentation.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, along with a portfolio-ready project file that can be showcased to employers or clients.

11. Primavera P6

Software Course | Duration: 5 Weeks

A project scheduling and control course using Primavera P6 — the construction industry standard for planning, tracking, and reporting on large-scale projects.

Who Should Enroll: Site engineers, planning engineers, and project managers responsible for construction scheduling and progress tracking.

Prerequisites

- Basic understanding of construction sequencing
- No prior scheduling software experience required

Learning Outcomes

- Set up projects, calendars, and work breakdown structures (WBS)
- Build activity networks with correct logic and dependencies
- Apply resource and cost loading to a schedule
- Run critical path method (CPM) analysis
- Track progress against baseline and generate management reports

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Project Setup	– Interface, project setup, and calendars – Creating WBS and activities
Week 2	Scheduling Logic	– Activity relationships and dependencies – Critical path method (CPM) scheduling
Week 3	Resources & Cost	– Resource and cost loading – Levelling and resource optimization
Week 4	Tracking & Reporting	– Baseline setup and progress tracking – Schedule updates, delay analysis, and reporting – Earned value basics
Week 5	Capstone Project	– Complete project schedule from WBS to tracked baseline – Progress report and S-curve generation

Tools & Reference Standards

- Primavera P6 Professional
- S-curve & EVM reporting techniques

Capstone Project

A complete, resource-loaded construction schedule with a tracked baseline, progress update, and S-curve report.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, along with a portfolio-ready project file that can be showcased to employers or clients.

12. Navisworks

Software Course | Duration: 4 Weeks

A BIM coordination course focused on aggregating multi-discipline models, running clash detection, and building 4D construction sequences — essential for avoiding costly site rework.

Who Should Enroll: BIM coordinators, structural/MEP engineers, and project engineers responsible for model coordination.

Prerequisites

- Basic familiarity with any BIM authoring tool (Revit, Tekla, etc.)

Learning Outcomes

- Aggregate models from multiple software formats into one coordination model
- Configure and run clash detection rules across disciplines
- Interpret and resolve clash reports collaboratively
- Build 4D construction sequences using Timeliner
- Perform basic quantification and simulation walkthroughs

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Model Aggregation	– Interface and model aggregation from multiple formats – Navigation and viewpoint management
Week 2	Clash Detection	– Clash detection setup and rule configuration – Running and reviewing clash test batches
Week 3	4D Scheduling	– Timeliner basics for 4D construction sequencing – Linking schedule activities to model objects
Week 4	Capstone Project	– Multi-discipline coordination model with clash detection report – 4D construction sequence simulation and walkthrough

Tools & Reference Standards

- Navisworks Manage
- Timeliner
- Multi-format model aggregation (Revit, IFC, DWG)

Capstone Project

A coordinated multi-discipline model with a full clash report and a 4D construction sequence simulation.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, along with a portfolio-ready project file that can be showcased to employers or clients.

2. Technical Courses – B.E. Civil Level

Core engineering theory and code-based design, from mechanics to project management

1. Engineering Mechanics

Technical Course | Duration: 4 Weeks

The foundational course in statics and basic structural behavior — building the analytical thinking every structural, geotechnical, and construction course afterward depends on.

Who Should Enroll: First/second-year civil engineering students and anyone needing a strong mechanics foundation before structural design courses.

Prerequisites

- Basic mathematics (trigonometry, vectors)

Learning Outcomes

- Analyze force systems and draw accurate free body diagrams
- Calculate centroid, center of gravity, and moment of inertia
- Analyze simple trusses using method of joints/sections
- Solve friction and simple-machine problems
- Apply mechanics principles to basic beam loading scenarios

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Fundamentals	– Force systems, equilibrium, and free body diagrams – Centroid and center of gravity
Week 2	Analysis	– Moment of inertia calculations – Analysis of trusses (method of joints/sections) – Friction and simple machines
Week 3	Design & Code-Based Problems	– Analysis of beams under basic loading conditions – Standard textbook and competitive-exam style problem sets
Week 4	Practical Application	– Application-based problem solving for real structural scenarios – Doubt-clearing and revision session

Tools & Reference Standards

- Manual calculation methods
- Basic mechanics reference tables

Capstone Project

A solved problem-set portfolio covering force analysis, trusses, and beam loading — usable as exam and interview preparation.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, backed by a solved design/analysis portfolio referencing relevant IS Codes.

2. Structural Analysis

Technical Course | Duration: 6 Weeks

A rigorous course in classical structural analysis methods — covering determinate and indeterminate structures — that forms the theoretical backbone for RCC, steel, and software-based design courses.

Who Should Enroll: Civil engineering students and design trainees preparing for structural design roles or GATE/competitive exams.

Prerequisites

- Engineering Mechanics or equivalent foundational knowledge

Learning Outcomes

- Differentiate determinate and indeterminate structural systems
- Analyze beams and frames using moment distribution and slope deflection methods
- Construct influence lines for moving loads on beams
- Analyze trusses and arches for various loading conditions
- Cross-verify manual analysis results against software output

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Fundamentals	– Types of structures, loads, and support conditions – Determinate vs indeterminate structures
Week 2–3	Analysis	– Analysis of beams and frames: moment distribution method – Slope deflection method – Influence lines for beams
Week 4	Design	– Analysis of trusses and arches
Week 5	Code-Based Problems	– IS code-referenced analysis problems and exam-style sets
Week 6	Practical Application	– Analysis of a real building frame using manual methods – Cross-checking results using structural software

Tools & Reference Standards

- Manual analysis methods
- Cross-verification using STAAD.Pro/ETABS (optional)

Capstone Project

A fully solved manual analysis of a real building frame, cross-verified against software output.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, backed by a solved design/analysis portfolio referencing relevant IS Codes.

3. RCC Design

Technical Course | Duration: 8 Weeks

A complete limit-state design course covering beams, slabs, columns, and footings as per IS 456 — the single most in-demand skill for structural design job roles.

Who Should Enroll: Civil engineering graduates and design trainees preparing for structural design engineer roles.

Prerequisites

- Structural Analysis (recommended)
- Basic understanding of concrete and steel properties

Learning Outcomes

- Apply limit state design principles as per IS 456
- Design singly and doubly reinforced beams for flexure, shear, and torsion
- Design one-way and two-way RC slabs
- Design isolated and combined column footings
- Produce a complete RC design set for a small building

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Fundamentals	– Concrete and steel properties, IS 456 basics – Limit state method concepts
Week 2–3	Beam Design	– Design of singly and doubly reinforced beams – Shear and torsion design
Week 4–5	Slab & Column Design	– Design of one-way and two-way slabs – Design of columns (short and slender)
Week 6	Footing Design	– Design of isolated and combined footings
Week 7	Code-Based Problems	– IS 456 & SP 16 based numerical problem sets
Week 8	Capstone Project	– Complete RC design of a small building: beam, slab, column, and footing – Design report compilation

Tools & Reference Standards

- IS 456, SP 16 design aids
- Manual and spreadsheet-based design calculations

Capstone Project

A complete RC design package (beam, slab, column, footing) for a small building, with a compiled design report.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, backed by a solved design/analysis portfolio referencing relevant IS Codes.

4. Steel Structure Design

Technical Course | Duration: 8 Weeks

A comprehensive steel design course as per IS 800 — covering tension/compression members, beam design, connections, and industrial roof trusses, preparing learners for steel design and detailing roles.

Who Should Enroll: Civil engineering graduates and design trainees targeting steel structure design and detailing careers.

Prerequisites

- Structural Analysis (recommended)
- Basic understanding of steel material properties

Learning Outcomes

- Design tension and compression members as per IS 800
- Design beams for flexure and shear including lateral stability checks
- Design bolted and welded connections
- Design column base plates and industrial roof trusses
- Deliver a complete steel design package for an industrial structure

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Fundamentals	- Properties of structural steel, IS 800 basics - Types of steel sections and connections
Week 2–3	Member Design	- Design of tension and compression members - Design of beams for flexure and shear
Week 4–5	Connections	- Design of bolted and welded connections - Design of column bases
Week 6	Trusses	Design of roof trusses for industrial structures
Week 7	Code-Based Problems	IS 800 based numerical design problem sets
Week 8	Capstone Project	- Design of a steel truss / industrial shed structure - Final design report compilation

Tools & Reference Standards

- IS 800 design aids
- Steel section tables (SP 6)

Capstone Project

Complete design of a steel truss or industrial shed structure, including connection and base plate design.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, backed by a solved design/analysis portfolio referencing relevant IS Codes.

5. Prestressed Concrete

Technical Course | Duration: 6 Weeks

A specialized design course covering prestressing concepts, loss calculations, and design of prestressed beams and slabs as per IS 1343 — a high-value skill for bridge and long-span structure design.

Who Should Enroll: Civil engineering graduates and structural design engineers interested in bridges and long-span structures.

Prerequisites

- RCC Design (recommended)

Learning Outcomes

- Differentiate pre-tensioning and post-tensioning systems
- Calculate stresses in prestressed sections at transfer and service
- Compute short-term and long-term losses in prestress
- Design prestressed beams and slabs as per IS 1343
- Analyze a prestressed girder/bridge deck element as a case study

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Fundamentals	- Concepts of prestressing: pre-tensioning vs post-tensioning - Materials used in prestressed concrete

Week	Focus	Topics Covered
Week 2–3	Analysis	– Analysis of stresses in prestressed sections – Losses in prestress: friction, anchorage, shrinkage, creep, relaxation
Week 4	Design	– Design of prestressed beams and slabs
Week 5	Code-Based Problems	– IS 1343 based numerical problem sets
Week 6	Practical Application	– Design case study of a prestressed girder / bridge deck element

Tools & Reference Standards

- IS 1343 codebook
- Manual and spreadsheet-based loss calculations

Capstone Project

A design case study of a prestressed girder or bridge deck element, including full loss calculations.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, backed by a solved design/analysis portfolio referencing relevant IS Codes.

6. Finite Element Analysis

Technical Course | Duration: 6 Weeks

An introduction to finite element theory and its application to structural elements — the conceptual foundation behind software like STAAD, ETABS, and SAFE.

Who Should Enroll: Civil engineering graduates and design engineers who want to understand what structural software is calculating behind the scenes.

Prerequisites

- Structural Analysis (recommended)
- Basic matrix algebra

Learning Outcomes

- Understand discretization and element types used in FEA
- Apply stiffness matrix method concepts to simple structures
- Formulate 1D and 2D elements and apply boundary conditions
- Apply FEA concepts to beams, plates, and frames
- Validate FEA-based modeling against software results

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Fundamentals	– Basics of discretization and element types – Stiffness matrix method concepts
Week 2–3	Analysis	– 1D and 2D element formulations – Boundary conditions and load application
Week 4	Design	– Application to beams, plates, and frames

Week	Focus	Topics Covered
Week 5	Code-Based Problems	– Numerical problems applying FEA principles
Week 6	Practical Application	– FEA-based modeling of a structural element with software-based validation

Tools & Reference Standards

- Manual matrix methods
- Software validation using STAAD/SAFE (introductory)

Capstone Project

A validated FEA-based model of a structural element, benchmarked against manual/software results.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, backed by a solved design/analysis portfolio referencing relevant IS Codes.

7. Surveying

Technical Course | Duration: 5 Weeks

A practical field-and-office surveying course covering traditional and modern instruments — essential groundwork for site engineers, planners, and estimators.

Who Should Enroll: Civil engineering students and site engineers who need field survey and site-planning skills.

Prerequisites

- None — suitable for beginners

Learning Outcomes

- Perform chain, compass, and leveling surveys accurately
- Conduct theodolite surveying and traversing
- Compute contours, areas, and volumes from survey data
- Understand total station and GPS surveying workflows
- Prepare a site plan from field survey data

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Fundamentals	– Chain and compass surveying basics – Leveling instruments and principles
Week 2	Analysis	– Theodolite surveying and traversing – Contouring and area/volume computation
Week 3	Design	– Total station and GPS surveying basics
Week 4	Code-Based Problems	– Survey computation and error correction problems
Week 5	Practical Application	– Field survey exercise and site plan preparation

Tools & Reference Standards

- Chain, compass, dumpy level

- Total station (introductory)
- Basic GPS survey concepts

Capstone Project

A complete site plan prepared from a hands-on field survey exercise.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, backed by a solved design/analysis portfolio referencing relevant IS Codes.

8. Building Materials & Concrete Technology

Technical Course | Duration: 5 Weeks

A materials-focused course covering cement, aggregates, admixtures, and concrete mix design — the quality-control foundation behind every durable construction project.

Who Should Enroll: Civil engineering students, site engineers, and QA/QC professionals.

Prerequisites

- None — suitable for beginners

Learning Outcomes

- Identify properties and grades of cement, aggregates, and admixtures
- Distinguish types and grades of concrete for different applications
- Design a concrete mix as per IS 10262
- Interpret workability and strength test results
- Apply durability and quality-control principles on site

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Fundamentals	– Properties of cement, aggregates, and admixtures – Types and grades of concrete
Week 2	Analysis	– Concrete mix design principles – Testing of materials: workability and strength
Week 3	Design	– Durability and quality control of concrete
Week 4	Code-Based Problems	– IS 10262 mix design based problem sets
Week 5	Practical Application	– Mix design exercise and lab test result interpretation

Tools & Reference Standards

- IS 10262 mix design method
- Standard lab testing procedures (slump, compressive strength)

Capstone Project

A complete concrete mix design worked from first principles, with interpreted lab test data.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, backed by a solved design/analysis portfolio referencing relevant IS Codes.

9. Construction Technology

Technical Course | Duration: 5 Weeks

A practical course on how buildings are actually built — from foundation to finishing — covering formwork, masonry, and modern construction methods used on active sites.

Who Should Enroll: Civil engineering students and site engineers who need practical construction sequencing knowledge.

Prerequisites

- None — suitable for beginners

Learning Outcomes

- Identify building components from foundation to roof
- Select appropriate construction materials and methods for a given scenario
- Understand formwork, scaffolding, and shoring systems
- Describe masonry and finishing work sequences
- Evaluate modern construction techniques and equipment choices

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Fundamentals	– Building components: foundation to roof – Construction materials and methods
Week 2	Analysis	– Formwork, scaffolding, and shoring systems – Masonry and finishing works
Week 3	Design	– Modern construction techniques and equipment
Week 4	Code-Based Problems	– Construction methodology-based case problems
Week 5	Practical Application	– Case study of a construction site sequence and methodology report

Tools & Reference Standards

- Site methodology case studies
- Standard construction sequencing references

Capstone Project

A methodology report analyzing the construction sequence of a real or case-study site.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, backed by a solved design/analysis portfolio referencing relevant IS Codes.

10. Geotechnical Engineering

Technical Course | Duration: 6 Weeks

A soil mechanics and foundation-readiness course covering classification, strength, and bearing capacity — critical knowledge before any foundation can be safely designed.

Who Should Enroll: Civil engineering graduates and design engineers preparing for foundation and geotechnical design roles.

Prerequisites

- Basic engineering mechanics

Learning Outcomes

- Classify soils and determine index properties
- Apply permeability and effective stress concepts
- Analyze compaction and consolidation behavior of soils
- Determine shear strength parameters of soils
- Calculate bearing capacity and assess basic slope stability

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Fundamentals	– Soil classification and index properties – Permeability and effective stress concepts
Week 2–3	Analysis	– Compaction and consolidation of soils – Shear strength of soils
Week 4	Design	– Bearing capacity and slope stability basics
Week 5	Code-Based Problems	– IS 1904 & IS 6403 based numerical problem sets
Week 6	Practical Application	– Soil investigation report interpretation and bearing capacity calculation

Tools & Reference Standards

- IS 1904, IS 6403 codebooks
- Standard soil test data interpretation

Capstone Project

A bearing capacity calculation and foundation-readiness assessment based on a real soil investigation report.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, backed by a solved design/analysis portfolio referencing relevant IS Codes.

11. Foundation Engineering

Technical Course | Duration: 6 Weeks

A design-focused course covering shallow and deep foundation systems — a direct follow-on to Geotechnical Engineering and RCC Design, and a core skill for structural design roles.

Who Should Enroll: Civil engineering graduates and structural/geotechnical design engineers.

Prerequisites

- Geotechnical Engineering and RCC Design (recommended)

Learning Outcomes

- Select appropriate foundation types based on site investigation data
- Design isolated and combined shallow foundations
- Design pile foundations for various soil and load conditions
- Understand raft/mat foundation design basics

- Address foundation design challenges on problematic soils

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Fundamentals	– Types of foundations and site investigation – Load transfer mechanisms
Week 2–3	Analysis	– Design of shallow foundations (isolated, combined) – Design of pile foundations
Week 4	Design	– Raft and mat foundation design basics – Foundations on problematic soils
Week 5	Code-Based Problems	– IS 456 & IS 2911 based design problems
Week 6	Practical Application	– Foundation design case study for a multi-storey building

Tools & Reference Standards

- IS 456, IS 2911 codebooks
- Manual and spreadsheet-based foundation design

Capstone Project

A complete foundation design case study for a multi-storey building, including pile/raft design as applicable.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, backed by a solved design/analysis portfolio referencing relevant IS Codes.

12. Quantity Surveying & Estimation

Technical Course | Duration: 5 Weeks

A commercially critical course covering measurement, rate analysis, and BOQ preparation — the skillset behind every accurate construction budget and tender document.

Who Should Enroll: Civil engineering graduates, site engineers, and aspiring quantity surveyors/estimators.

Prerequisites

- Ability to read building drawings (AutoCAD course recommended)

Learning Outcomes

- Read construction drawings for accurate quantity take-off
- Estimate earthwork, concrete, and masonry quantities
- Perform rate analysis for common building items
- Prepare a professional bill of quantities (BOQ)
- Apply IS 1200 measurement standards to real project drawings

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Fundamentals	– Principles of estimation and units of measurement – Reading drawings for quantity take-off

Week	Focus	Topics Covered
Week 2–3	Analysis	– Estimation of earthwork, concrete, and masonry – Rate analysis for building items
Week 4	Design	– Preparation of bill of quantities (BOQ)
Week 4b	Code-Based Problems	– IS 1200 based measurement problem sets
Week 5	Practical Application	– Full quantity estimation and BOQ preparation for a sample building

Tools & Reference Standards

- IS 1200 measurement code
- Spreadsheet-based estimation templates

Capstone Project

A complete quantity estimation and BOQ for a sample building, ready for tendering use.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, backed by a solved design/analysis portfolio referencing relevant IS Codes.

13. Construction Planning & Management

Technical Course | Duration: 5 Weeks

A project management course covering scheduling techniques, resource planning, and cost control — the managerial companion to Primavera P6 for anyone running or overseeing construction projects.

Who Should Enroll: Civil engineering graduates, site engineers, and aspiring project/planning engineers.

Prerequisites

- None — suitable for beginners; pairs well with Primavera P6

Learning Outcomes

- Understand the construction project life cycle and organization structures
- Build a work breakdown structure (WBS) for a project
- Apply bar chart, CPM, and PERT scheduling techniques
- Plan and allocate resources across a construction schedule
- Apply basic cost control and budgeting principles

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Fundamentals	– Project life cycle and organization structures – Work breakdown structure (WBS) basics
Week 2–3	Analysis	– Bar charts, CPM, and PERT scheduling techniques – Resource planning and allocation
Week 4	Design	– Cost control and budgeting basics
Week 4b	Code-Based Problems	– Scheduling and resource-leveling numerical problems

Week	Focus	Topics Covered
Week 5	Practical Application	– Construction schedule and resource plan for a sample project

Tools & Reference Standards

- CPM/PERT scheduling techniques
- Spreadsheet-based planning templates
- Primavera P6 (complementary)

Capstone Project

A complete construction schedule and resource plan for a sample project, with a basic cost control framework.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, backed by a solved design/analysis portfolio referencing relevant IS Codes.

14. BIM & Digital Construction

Technical Course | Duration: 5 Weeks

A modern digital-construction course covering BIM standards, multi-disciplinary coordination, and 4D/5D integration — the direction the entire construction industry is moving toward.

Who Should Enroll: Civil/architecture graduates and BIM-curious professionals who want to understand coordination workflows beyond single-software modeling.

Prerequisites

- Basic exposure to Revit or another BIM authoring tool is helpful

Learning Outcomes

- Explain BIM concepts and levels of development (LOD)
- Understand BIM collaboration workflows across disciplines
- Coordinate multi-disciplinary models and identify clashes
- Understand 4D scheduling and 5D cost integration basics
- Review a coordinated BIM model against standard documentation practices

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Fundamentals	– Introduction to BIM concepts and levels of development (LOD) – BIM workflow and collaboration basics
Week 2	Analysis	– Multi-disciplinary model coordination – Clash detection fundamentals
Week 3	Design	– 4D scheduling and 5D cost integration basics
Week 4	Code-Based Problems	– BIM standards and documentation-based exercises
Week 5	Practical Application	– Coordinated BIM model review and clash report for a sample project

Tools & Reference Standards

- Navisworks (coordination)
- BIM standards & LOD frameworks

Capstone Project

A coordinated multi-disciplinary BIM review with a clash report for a sample project.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, backed by a solved design/analysis portfolio referencing relevant IS Codes.

